Legalization & Cannabis Use

Stats 115 Final Project

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Our Research Question

- Was early state-level marijuana legalization in 2012 (Washington, Colorado) and 2014 (Alaska, Oregon) associated with higher marijuana-use rates among ages 18–25, beyond the broader national trend?

Background & Motivation

Why should we care about this topic?

- Marijuana legalization is a policy issue in the United States that goes beyond red/blue politics, as it reflects family dynamics, community experiences, and personal values.
- A key concern is whether legalization is associated with significantly higher marijuana use, especially among young adults.

Who will these results help in the future?

- Policymakers can use these results to better understand how marijuana regulation can affect younger demographics.
- Public health groups and schools can use this information to better educate and improve prevention efforts.
- Individuals can gain knowledge to create more informed opinions.

Dataset Overview

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Dimensions: 867 observations × 53 variables
Years: 2002 - 2018
States: 51 (includes Washington D.C.)
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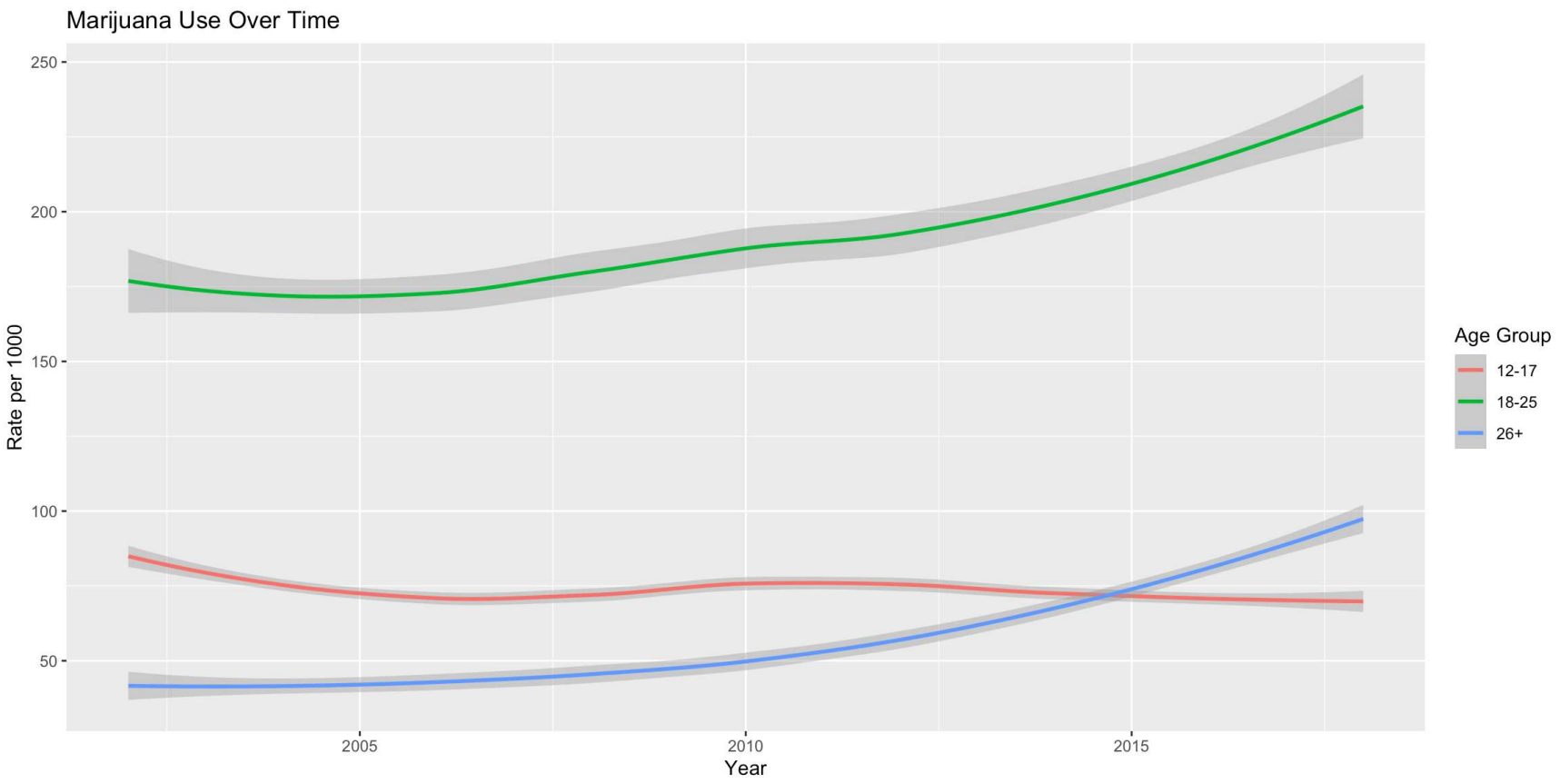
Key Variables:

- **Demographics**: State, Year, Population by age group
- **Substances**: Marijuana, Tobacco, Alcohol, Cocaine
- **Metrics**: Totals (counts) and Rates (per 1000 people)
- **Time frames**: Past month, past year usage

Age Groups:

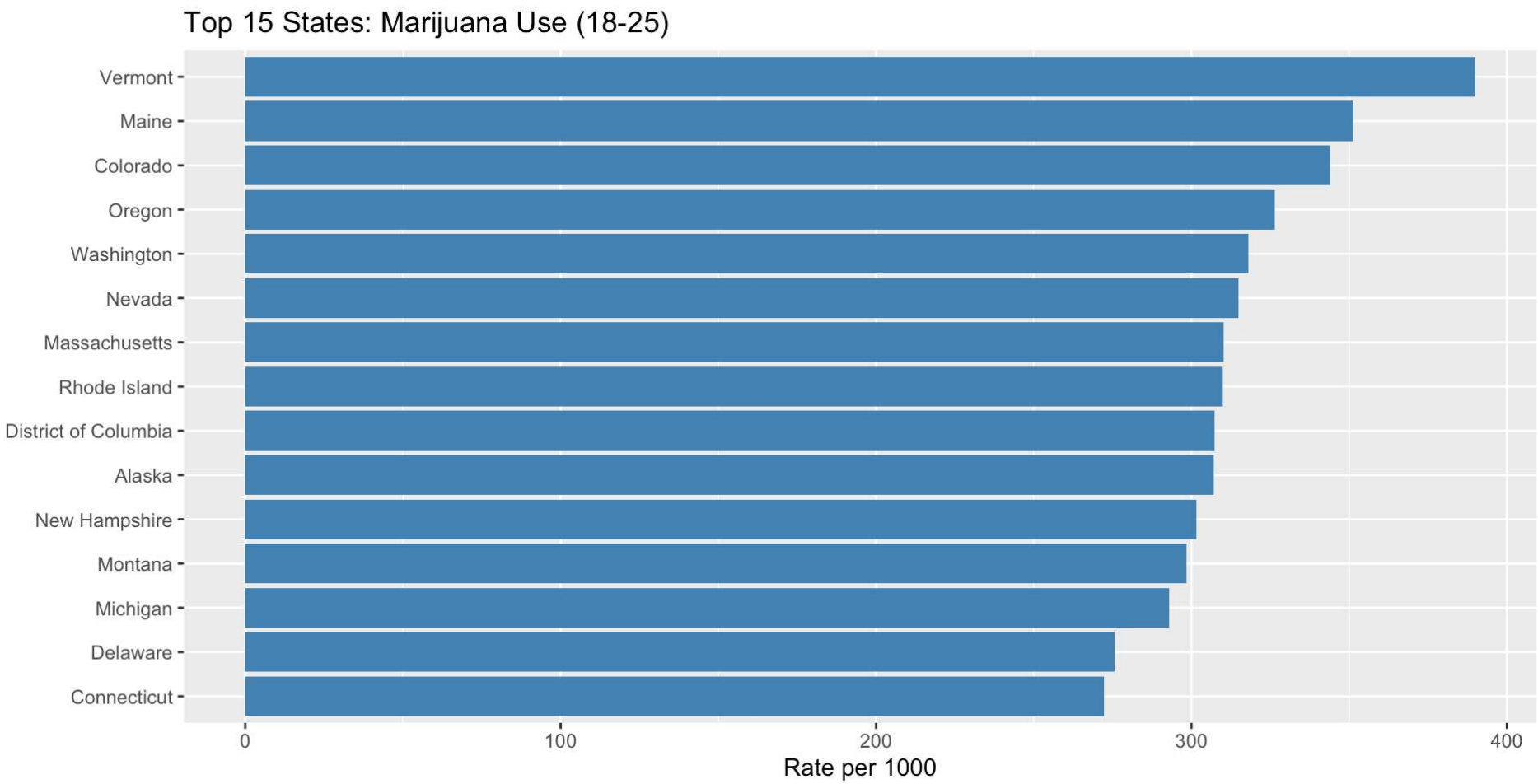
- 12-17 (adolescents)
- 18-25 (young adults)
- 26+ (adults)

Marijuana Trends by Age

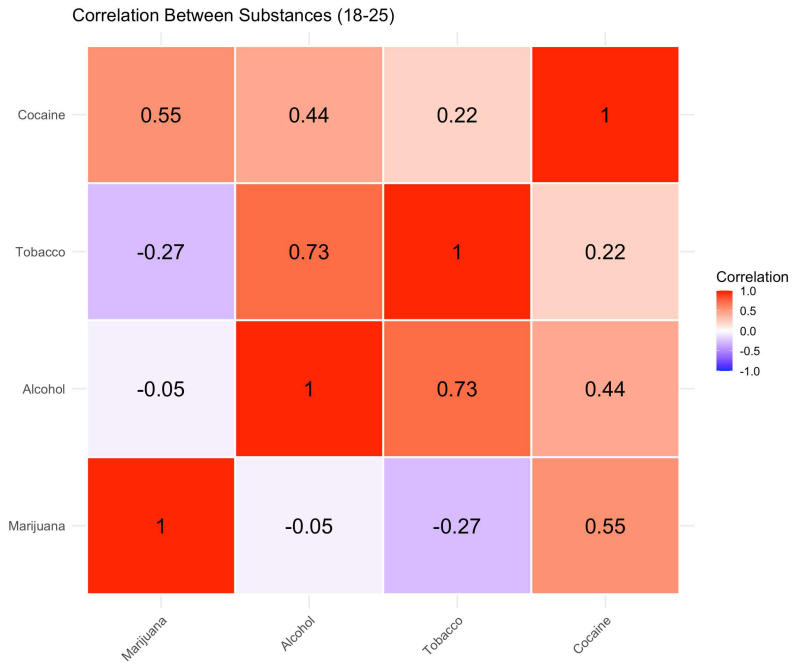


The 18–25 age group shows an increasing rate over time and remains significantly higher than the other two age groups.

States Ranked by Marijuana Use (2018)



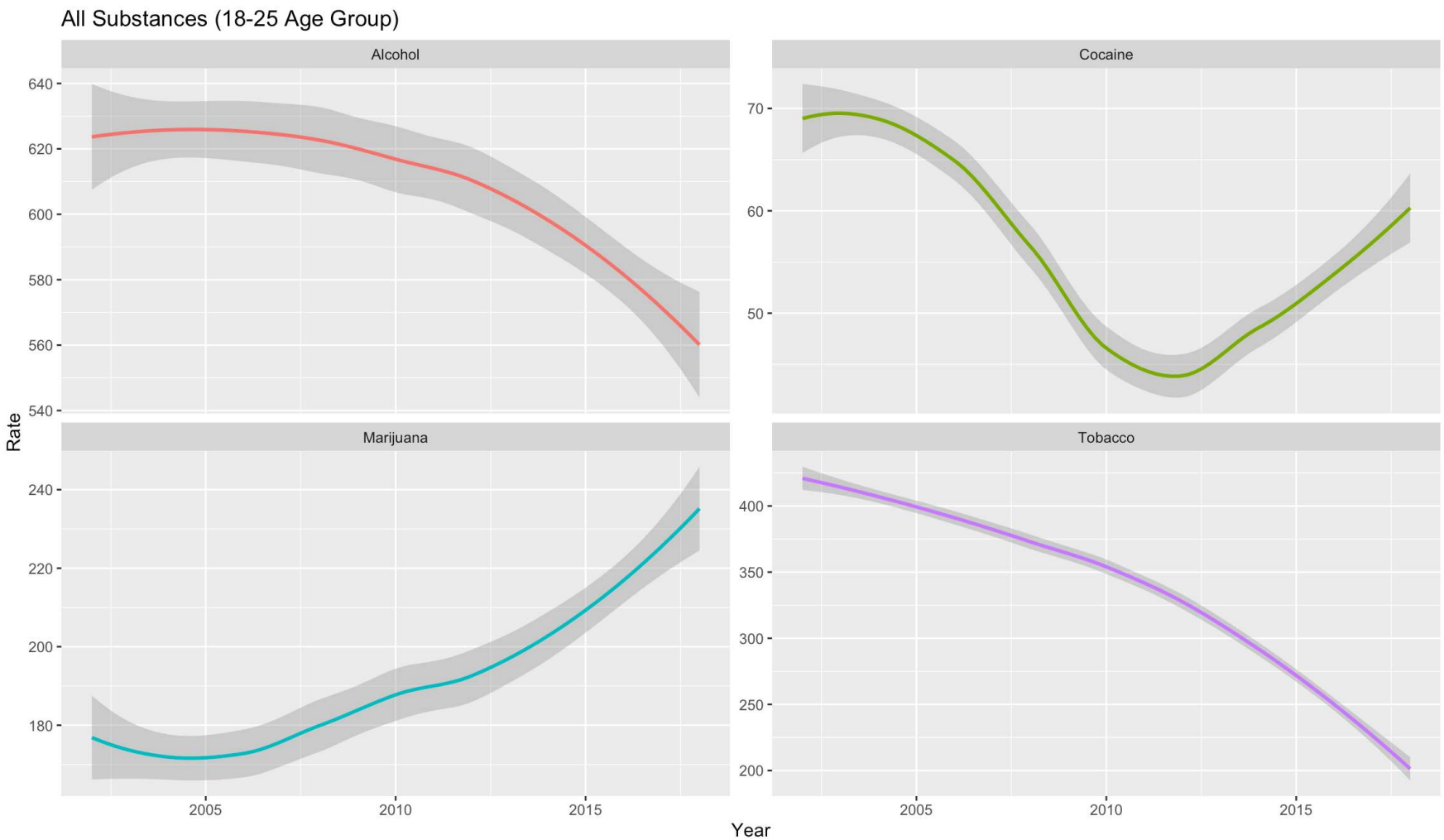
Correlation Matrix



Two distinct camps: Cocaine & Marijuana users (r = 0.55) vs Alcohol & Tobacco users (r = 0.73)

Key insight: Marijuana and Tobacco negatively correlated (r = -0.27) - both are smoking activities, individuals choose one over the other

All Substance Trends



Revisiting our Question

- Was early state-level marijuana legalization in 2012 (Washington, Colorado) and 2014 (Alaska, Oregon) associated with higher marijuana-use rates among ages 18–25, beyond the broader national trend?

Our exploratory analysis suggested that marijuana use among **18–25-year-olds** rose in the years following legalization. We then asked whether early-adopter states experienced an additional post-legalization increase relative to the broader national pattern, or whether usage was simply rising along with the national trajectory.

Legalization Timeline

Early Adopter States:

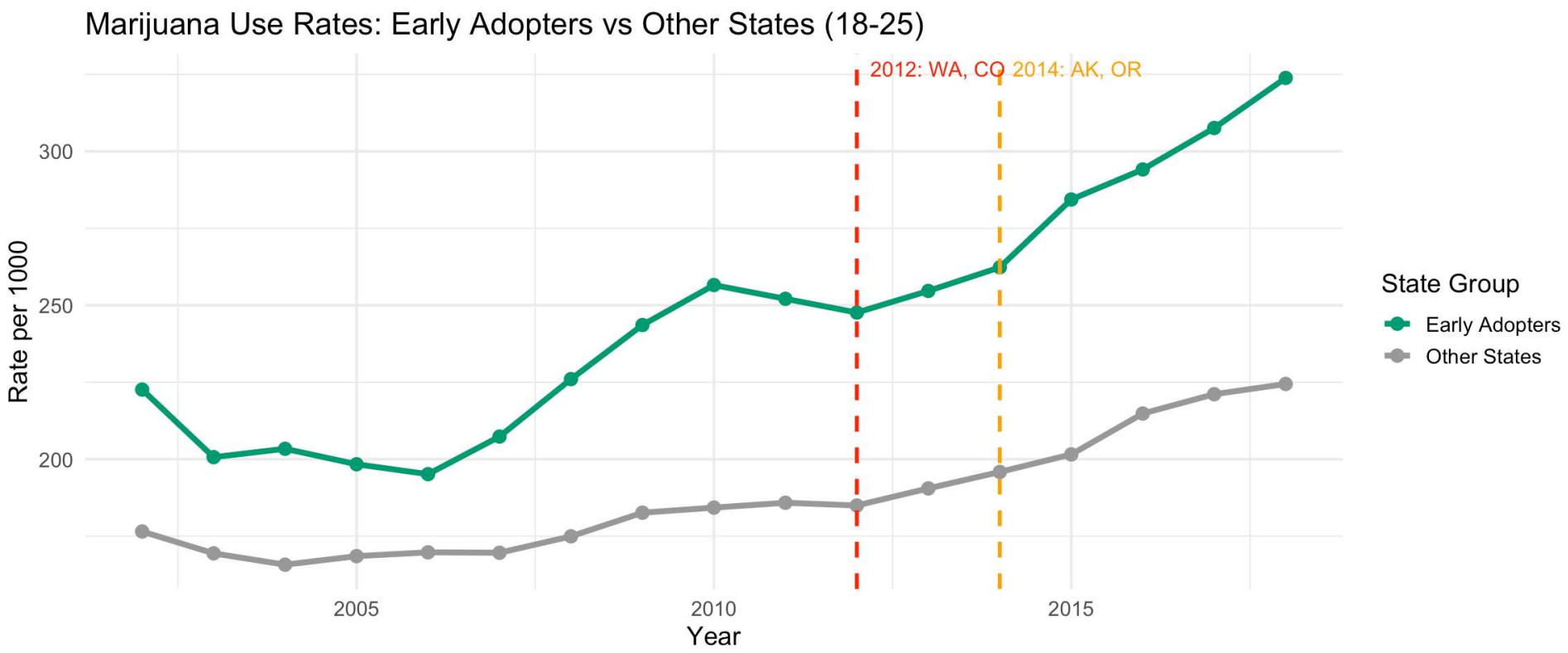
- **2012**: Washington & Colorado legalized recreational marijuana
- **2014**: Alaska & Oregon legalized recreational marijuana

Our Analysis:

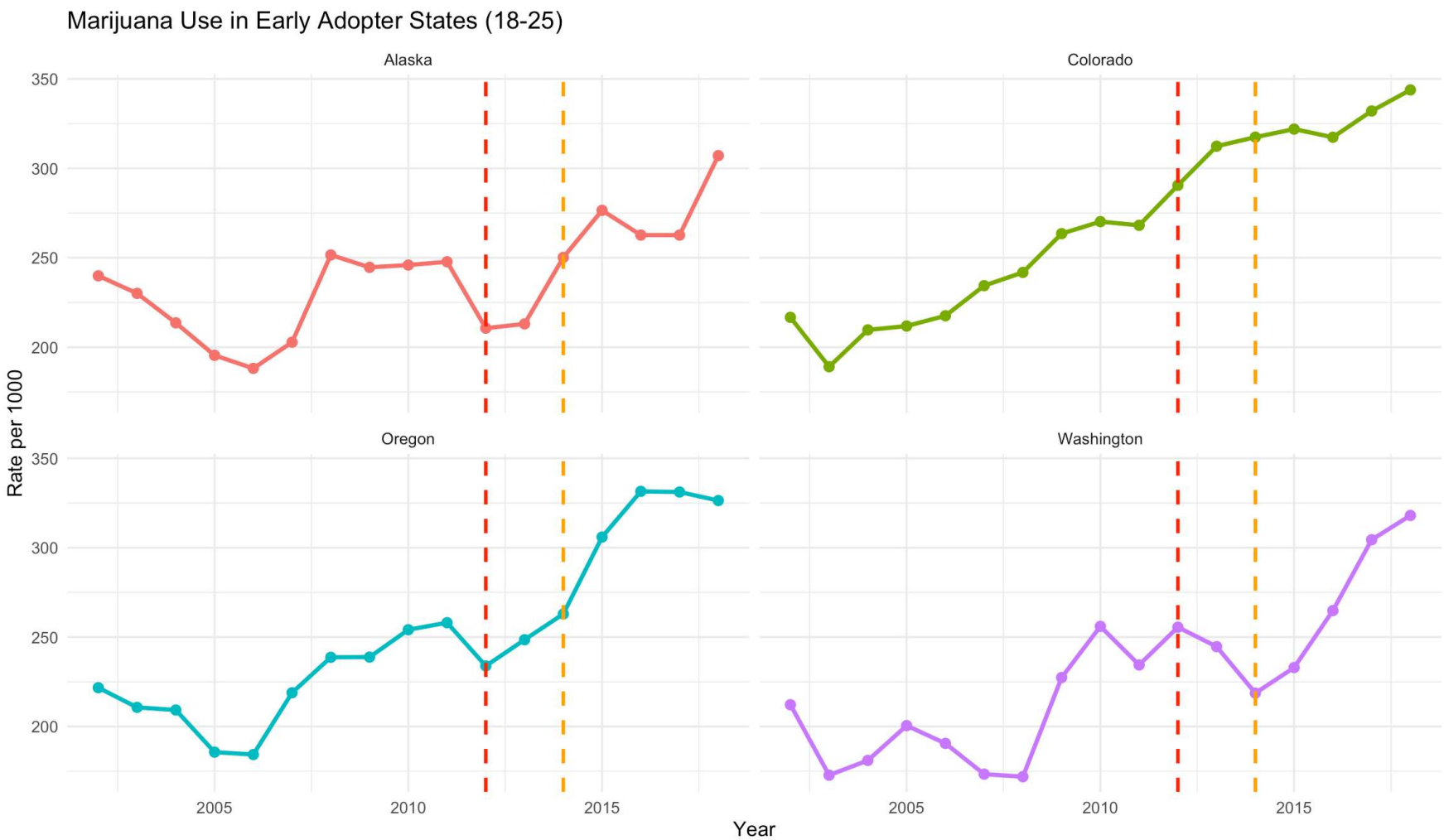
Compare marijuana usage rates in these 4 early-adopter states vs. all other states to identify:

1. Pre-legalization baseline differences
2. Post-legalization divergence (if any)
3. Whether early-adopter states showed larger post-legalization increases than other states

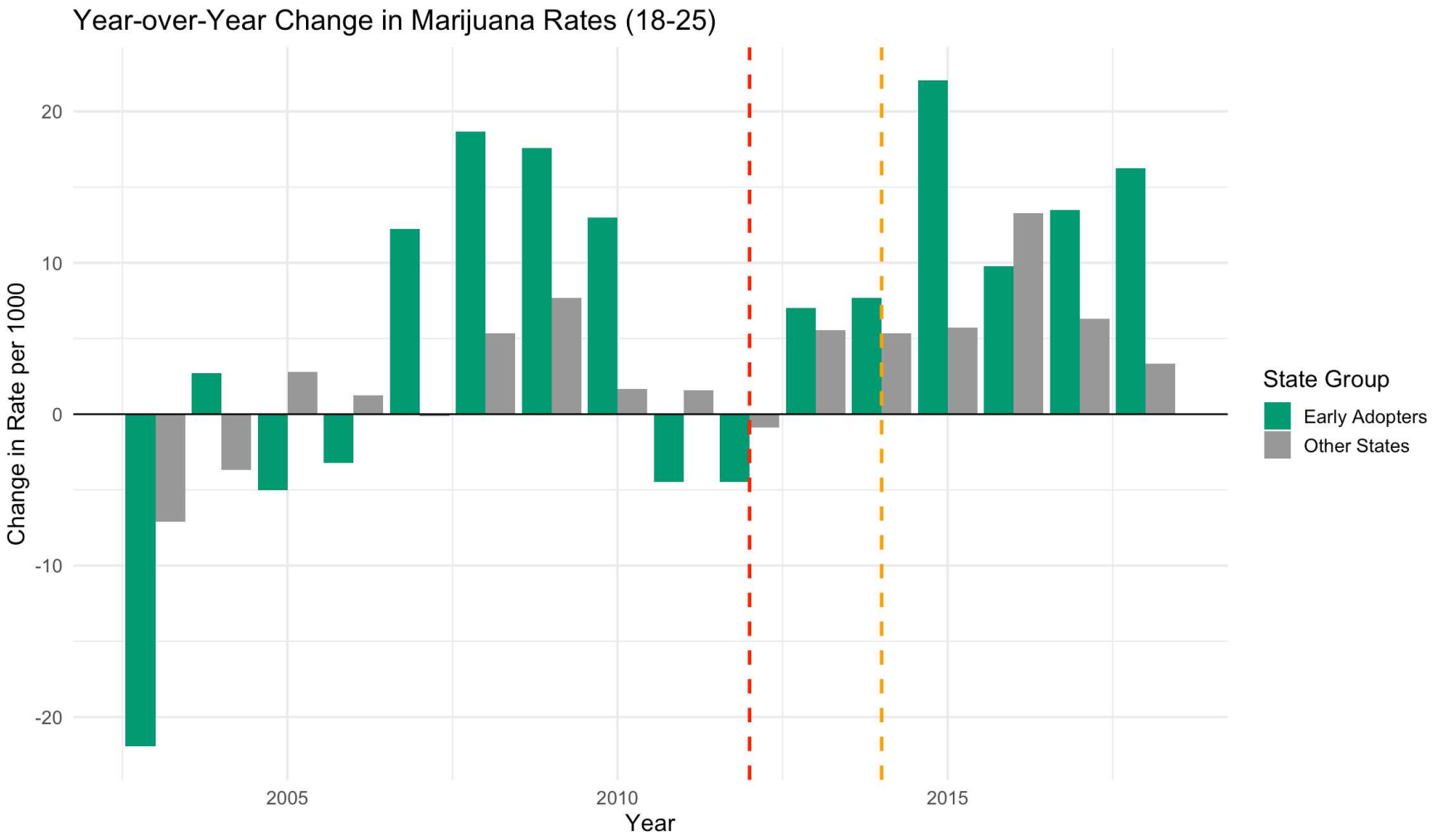
Marijuana Rates: Early Adopters vs Others



Pre vs Post Legalization Trends



Rate of Change Analysis



Findings

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Pre-2012 Average Rates (per 1000):
  Early Adopters: 220.58
  Other States: 174.73
  Difference: 45.85
Post-2014 Average Rates (per 1000):
  Early Adopters: 294.44
  Other States: 211.56
  Difference: 82.88
Change from Pre-2012 to Post-2014:
  Early Adopters: + 73.86 ( 33.5 % increase)
  Other States: + 36.83 ( 21.1 % increase)
```

Note: This analysis uses all pre-2012 vs all post-2014 data for comprehensive coverage.

Interpretation:

- Larger increases in early-adopter states are consistent with a positive post-legalization association, but do **not establish causality by themselves**
- Similar increases across groups would be more consistent with a broader national trend
- Baseline differences suggest early-adopter states may already have had stronger underlying demand or different pre-existing preferences

Initial Summary

Pre-legalization (before 2012):

- Early adopters: 220.6 per 1000
- Other states: 174.7 per 1000
- Early-adopter states already had substantially higher marijuana-use rates before legalization

Post-legalization (2014+):

- Early adopters: 294.4 per 1000
- Other states: 211.6 per 1000
- The gap widened further in the post-legalization period

Key Pattern:

- Early adopters: **33.5% increase** (+73.9 per 1000)
- Other states: **21.1% increase** (+36.8 per 1000)

Important Caution

Because early-adopter states already had higher baseline usage and appear to have been increasing faster even before legalization, this comparison does **not establish a causal effect by itself**.

The post-legalization divergence is consistent with legalization being associated with higher marijuana-use rates, but part of the difference may also reflect pre-existing state-level preferences and trends.

Proper Interpretation: Early-adopter states experienced larger increases in marijuana use among ages 18–25 than other states, but the evidence here should be interpreted as **associational rather than definitively causal**. Legalization may have contributed to the increase, but these states were already different before legalization.

Statistical Testing: Did Legalization Matter?

Modeling Strategy:

Because legalization occurred at different times across the treated states, we define a **state-specific post-legalization indicator**:

- $Post_{it} = 1$ if state i is in a legalized period in year t
- $Post_{it} = 0$ otherwise

This means $Post_{it}$ already represents the treatment exposure for early-adopter states after legalization. It is effectively a **treated × post** indicator, so we do **not** need a separate interaction term such as $EarlyAdopter_i \times Post_t$ in this model.

Interpretation Goal

We estimate whether marijuana-use rates in ages 18–25 were higher in treated states **after their legalization year**, while also accounting for:

- year-specific national effects shared across states
- persistent baseline differences across states

Two-Sample t-test on Growth Rates

Let's first run a simple t-test:

$$H_0 : \mu_{\text{Early growth}} = \mu_{\text{Other growth}}$$

$$H_1 : \mu_{\text{Early growth}} > \mu_{\text{Other growth}}$$

```
Two-Sample t-test: Early Adopters vs Other States
Mean growth Early Adopters: 34.9
Mean growth Other States: 23.13
t-statistic: 0.85
p-value: 0.2249

Conclusion: No statistically significant difference in average growth was detected
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Note: This t-test uses 2010-2011 (pre) vs 2015-2016 (post) for a clean comparison around the legalization dates.

A Bayesian Approach

Why Bayesian?

- Quantify **uncertainty** in the effect of legalization
- Get **probability distributions** instead of just point estimates
- Make probabilistic statements such as: “What is the probability that the post-legalization effect is positive?”
- Incorporate **prior knowledge** about drug use trends

Proposed Model

$$Rate_{it} \sim \text{Normal}(\mu_{it}, \sigma)$$

$$\mu_{it} = \alpha_i + \lambda_t + \beta Post_{it}$$

Where:

- $Rate_{it}$ is the marijuana-use rate for state i in year t
- α_i is the state-specific baseline level
- λ_t is the year-specific effect shared by all states
- $Post_{it}$ is a state-specific post-legalization indicator

Why use year effects instead of a linear year trend?

A single linear time term assumes marijuana-use rates changed at a constant national rate over time. Using year effects is more flexible: it allows each year to have its own national shift, which better captures non-linear changes in marijuana use over the sample period.

Prior Choices and Statistical Decisions

Weakly informative priors used:

- Intercept: $Normal(180, 50)$
- Coefficients ($Post_{it}$ and year effects): $Normal(0, 20)$
- Residual SD: $Exponential(1)$

They reflect plausible rate levels, keep effects centered at no change unless the data suggest otherwise, and help regularize the model without overwhelming the evidence from the data.

Important Note

This is **not** a slope-interaction model such as

$$Rate \sim Year + Group + Year \times Group$$

Instead, the fitted model estimates a **post-legalization level shift** through $Post_{it}$, while allowing each state to have its own baseline level through the random intercept α_i.

Model Interpretation

Key Parameter: $\beta\,(Post_{it})$

If $\beta > 0$, then marijuana-use rates were higher in treated states during their post-legalization years, after accounting for:

- baseline differences across states through α_i
- nationwide year-to-year shocks through λ_t

Because $Post_{it}$ is defined separately for each state based on its own legalization year, it already serves as the treatment indicator for "treated state after legalization."

Parameter meanings:

- α_i: state-specific baseline marijuana-use level
- λ_t: year-specific national effect
- β: additional post-legalization shift in treated states

Model Interpretation Continued

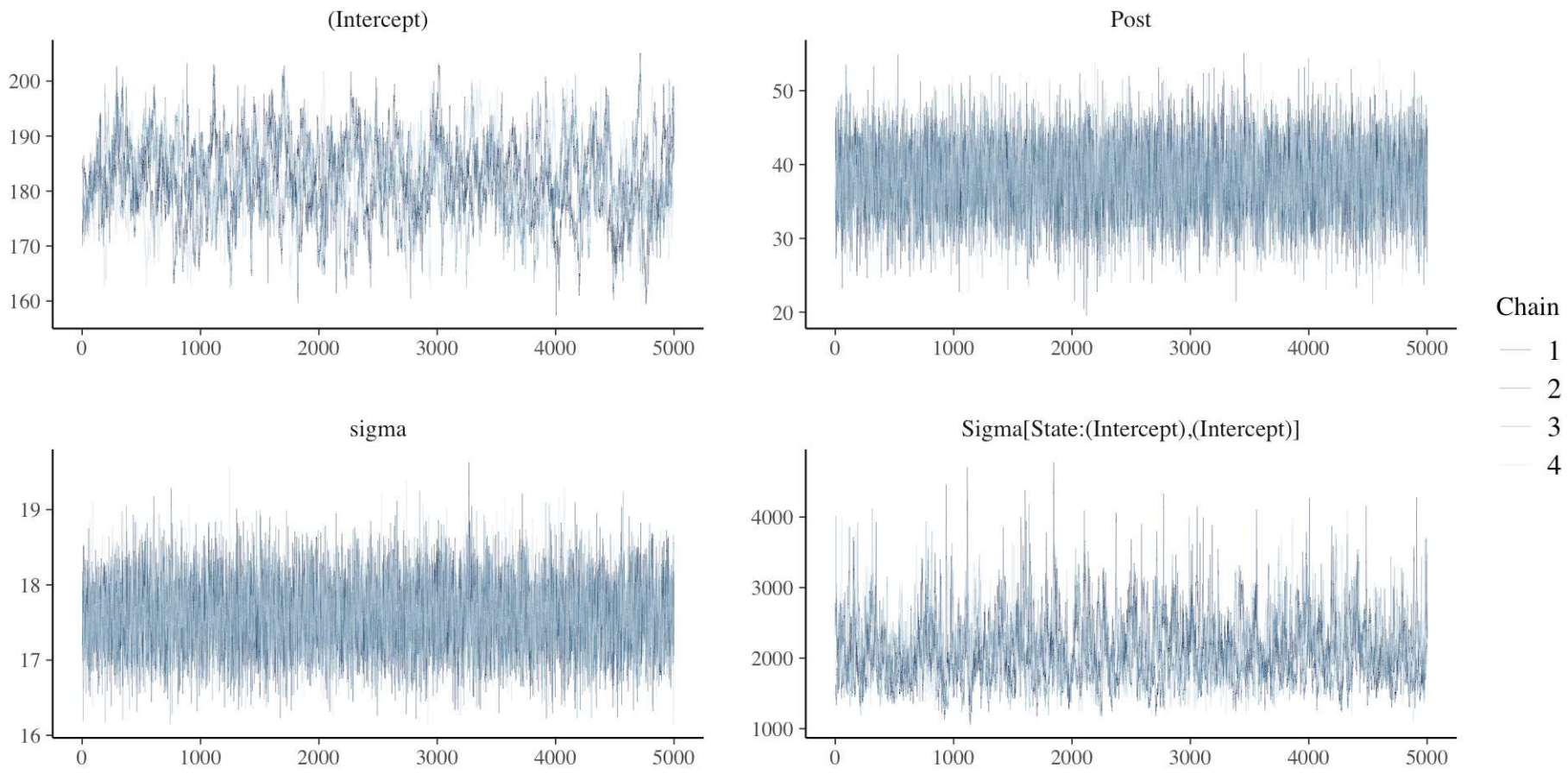
Model Note:

This specification is more flexible than a linear time trend because it does not force national marijuana-use rates to move in a straight line over time.

Posterior Predictive Checks:

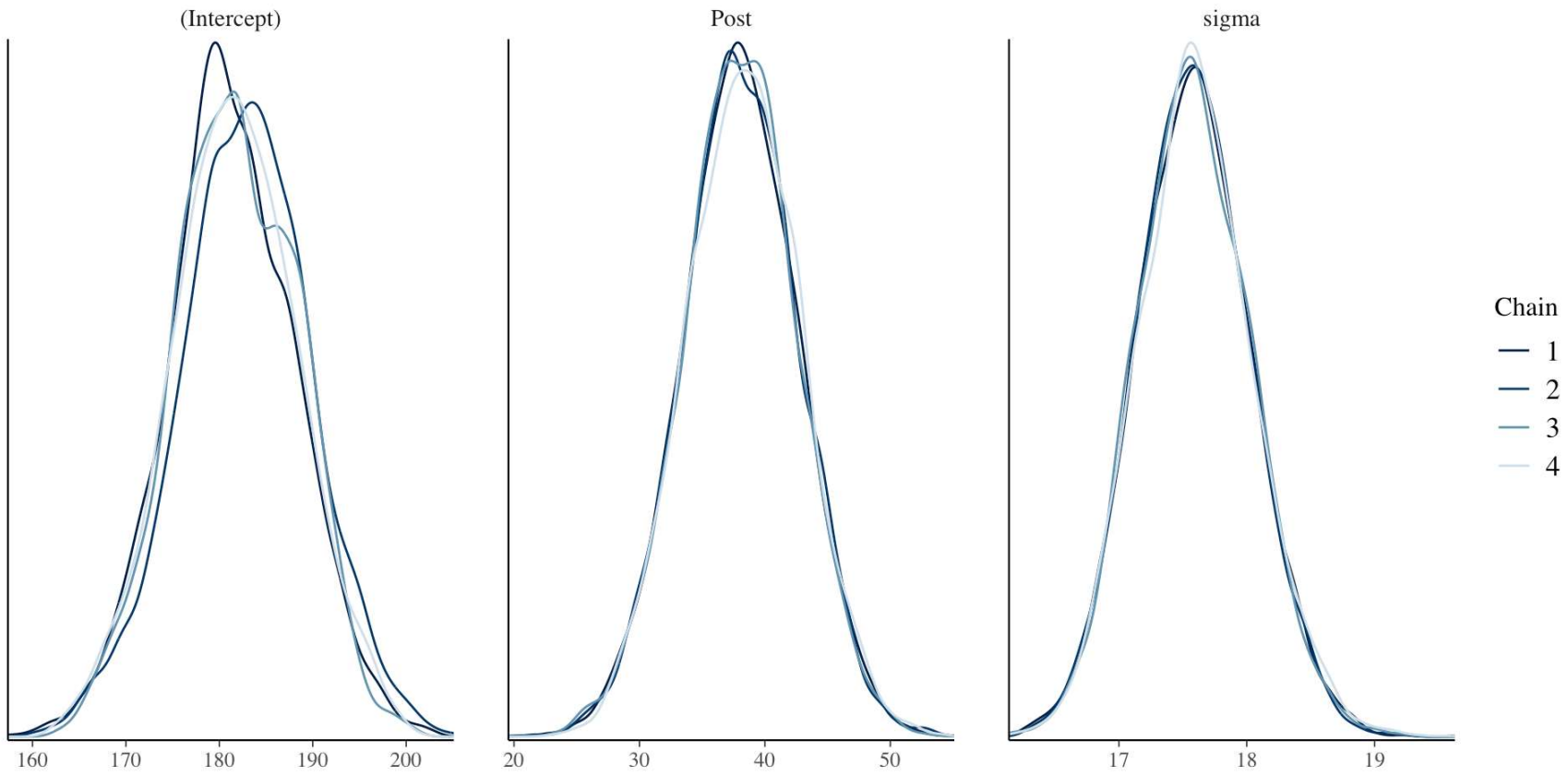
- Simulate replicated state-year rates from the posterior
- Compare simulated values to observed trends over time
- Check whether the model captures national year effects, state-level variation, and post-legalization shifts

MCMC Trace



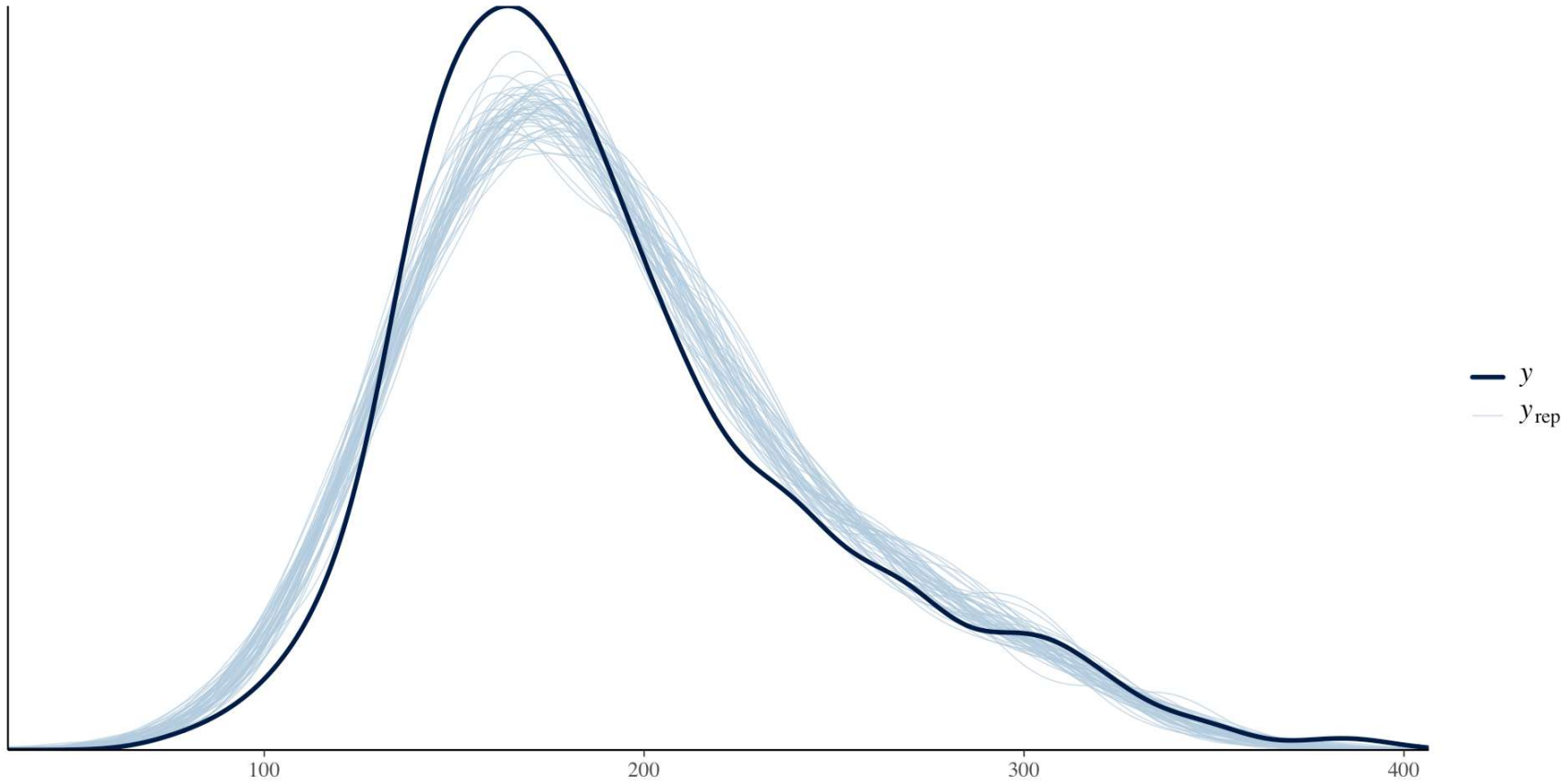
The traceplots show good mixing across chains

MCMC Density Overlay



Density overlays are quite consistent across chains. Intercept chains showed slight variation, but overall the posterior simulation converged well and the MCMC samples are reliable for inference.

Posterior Predictive Check



Our simulated data from the posterior match the observed distribution and trend quite well. The model captures the main structure of the marijuana-use data reasonably well.

Bayesian Analysis Results

Posterior Summary:

- 95% credible interval for β ($Post_{it}$): $[29.02,\ 46.99]$
- $P(\beta > 0 \mid data) = 1.00$

Interpretation: After accounting for state-level baseline differences and year-specific national effects, the estimated post-legalization coefficient remains clearly positive. Under this model, post-legalization years in treated states were associated with an additional increase of about 29 to 47 rate points in marijuana-use rates among ages 18–25. This estimate reflects a post-legalization level shift rather than a different slope over time.

Comparing the Approaches

Descriptive Analysis:

- Early-adopter states showed larger increases in marijuana-use rates than other states
- However, these states also had higher baseline usage before legalization

Frequentist Test:

- The two-sample t-test found higher average growth in early-adopter states
- However, the difference was not statistically significant

Bayesian Model:

- The model controls for state-level baseline differences and year-specific national effects
- The post-legalization indicator $Post_{it}$ remained strongly positive: 95% credible interval $[29.02,\ 46.99]$

Comparing the Approaches Continued

Key Insight:

Across all three approaches, early-adopter states showed larger post-legalization increases in marijuana-use rates among ages 18–25. The Bayesian model provides the clearest evidence of a positive post-legalization **association**, even after accounting for persistent state differences and flexible year-to-year national shifts. At the same time, because early-adopter states already differed before legalization, the result should be interpreted as strongly suggestive rather than definitively causal.

Where could we go from here?

- We could study other factors like dispensary access, advertising, and explore different demographics outside of just age.
- We could increase the timeframe we are looking at to study long term effects, and potentially combine our findings with other datasets that include information on post 2018 legalization.
- We could collect more detailed data, such as county-level, instead of only state-level averages.
- We could explore the link between marijuana and other substances, studying more how their rates are related.

Thank you